

Applied Statistical Analysis of Employee Salaries Using Hypothesis Testing and Confidence Intervals

CMP-L011-0

Mathematics for Data Science

MSc Data Science

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TASK 1: PROBLEM SETUP

This analysis examines employee compensation patterns across two datasets to determine whether observed salaries differ significantly from established benchmarks.

1.1 Gender Pay Gap Dataset

Research Question

Does the average base salary of female employees differ significantly from the benchmark of £75,000?

Hypotheses

H₀: $\mu = 75,000$

The mean base salary of female employees equals £75,000.

H₁: $\mu \neq 75,000$

The mean base salary of female employees does not equal £75,000.

Population Parameter

μ = True mean base salary of female employees in the organization.

Test Selection and Justification

Selected Test: One-sample t-test for the mean

Justification:

1. Population standard deviation (σ) is unknown
2. Large sample size ($n > 30$) enables CLT
3. Two-tailed test
4. Significance level $\alpha = 0.05$

TASK 1: PROBLEM SETUP

1.1 GENDER PAY GAP DATASET

RESEARCH QUESTION:

Does the average base salary of female employees differ significantly from the benchmark of £75,000?

HYPOTHESES:

H_0 (Null Hypothesis): $\mu = 75,000$
The mean base salary of female employees equals £75,000

H_1 (Alternative Hypothesis): $\mu \neq 75,000$
The mean base salary of female employees does not equal £75,000

POPULATION PARAMETER:

μ = True mean base salary of female employees in the organization

TEST SELECTION:

One-sample t-test for the mean

JUSTIFICATION:

1. Population standard deviation (σ) is UNKNOWN
Must use sample standard deviation (s) as estimator
2. Large sample size expected ($n > 30$)
Central Limit Theorem applies
Sampling distribution of $\bar{x}$ is approximately normal
3. Two-tailed test
Testing for difference in EITHER direction (higher OR lower)
4. Significance level: $\alpha = 0.05$
5% chance of Type I error (rejecting true H_0)

TEST STATISTIC FORMULA:

$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$

CONFIDENCE INTERVAL FORMULA:

$CI = \bar{x} \pm t^* \times (s / \sqrt{n})$

1.2 HR DATASET

RESEARCH QUESTION:

Is the average employee salary significantly different from £70,000?

HYPOTHESES:

H_0 (Null Hypothesis): $\mu = 70,000$
The mean employee salary equals £70,000

H_1 (Alternative Hypothesis): $\mu \neq 70,000$
The mean employee salary does not equal £70,000

POPULATION PARAMETER:

μ = True mean salary of all employees in the organization

TEST SELECTION:

One-sample t-test for the mean (same justification as 1.1)

1.2 HR Dataset

Research Question

Is the average employee salary significantly different from the £70,000 benchmark?

Hypotheses

$H_0: \mu = 70,000$

The mean employee salary equals £70,000.

$H_1: \mu \neq 70,000$

The mean employee salary does not equal £70,000.

Population Parameter

μ = True mean salary of all employees.

Test Selection

Same methodology as 1.1.

TASK 2: ASSUMPTIONS AND ANALYSIS PLAN

Statistical inference requires specific assumptions about data collection and distribution properties.

TASK 2: ASSUMPTIONS AND ANALYSIS PLAN

This section examines the statistical assumptions underlying our hypothesis tests and discusses their validity for the Gender Pay Gap and HR datasets.

2.1 RANDOM SAMPLING ASSUMPTION

REQUIREMENT:

The sample must be randomly selected from the target population to ensure unbiased inference.

ASSESSMENT:

Both datasets are secondary data from organizational employee records. While we cannot verify true probabilistic random sampling, the data characteristics suggest reasonable representativeness:

Gender Pay Gap Dataset:

- 1,000 total employees across diverse characteristics
- Multiple job titles, departments, education levels, seniority levels
- 468 female employees included in our analysis
- Broad organizational coverage supports representativeness

HR Dataset:

- 311 employee records with varied organizational roles
- Multiple departments, positions, tenure levels
- Comprehensive demographic and performance attributes

POTENTIAL LIMITATIONS:

If systematic exclusions exist (e.g., certain departments, remote workers, part-time employees omitted), generalizability may be limited to the observed sample only.

CONCLUSION:

Assumption reasonably satisfied for organizational-level inference. Caution needed if generalizing beyond this specific organization.

2.2 INDEPENDENCE OF OBSERVATIONS

REQUIREMENT:

Each employee's salary must be independent – one employee's salary should not systematically influence another's beyond common organizational factors.

ASSESSMENT:

Some minimal dependence may exist due to:

- Common departmental pay scales and salary bands
- Shared managers influencing compensation decisions
- Organizational-wide salary adjustment policies
- Market-driven compensation changes

However, with large sample sizes ($n=468$ and $n=311$), minor dependencies have negligible effect on hypothesis test validity.

CONCLUSION:

Assumption reasonably satisfied. Individual salary records treated as independent observations.

2.3 NORMALITY OF THE SAMPLING DISTRIBUTION

REQUIREMENT:

The sampling distribution of the sample mean ($\bar{x}$) must be approximately normal for t-test validity.

CENTRAL LIMIT THEOREM (CLT):

Regardless of population distribution shape, the sampling distribution of $\bar{x}$ approaches normality as sample size increases. Generally, $n > 30$ is sufficient.

ASSESSMENT:

Gender Pay Gap Dataset:

- Sample size: $n = 468 \gg 30$ ✓
- CLT strongly applies

HR Dataset:

- Sample size: $n = 311 \gg 30$ ✓
- CLT strongly applies

Both samples greatly exceed the threshold, ensuring approximate normality of the sampling distribution via CLT, even if underlying salary distributions are skewed.

CONCLUSION:

Assumption strongly satisfied. CLT guarantees normality of the sampling distribution, making the t-test robust and appropriate.

2.4 LARGE-SAMPLE CONDITIONS AND TEST SELECTION

SAMPLE SIZE IMPLICATIONS:

With $n > 30$:

- CLT ensures normality of sampling distribution
- t-distribution closely approximates standard normal distribution
- Sample standard deviation (s) reliably estimates population σ

The t-test is preferred over the z-test because it is more conservative (wider critical values) and properly accounts for uncertainty in estimating σ when population standard deviation is unknown.

DEGREES OF FREEDOM:

- Gender Pay Gap: $df = 468 - 1 = 467$
- HR Dataset: $df = 311 - 1 = 310$

CONCLUSION:

Large-sample conditions satisfied; t-test is appropriate.

2.5 SIGNIFICANCE LEVEL AND DECISION RULE

SIGNIFICANCE LEVEL: $\alpha = 0.05$

This represents a 5% probability of Type I error (rejecting a true null hypothesis), corresponding to 95% confidence. Standard in business research.

DECISION RULE FOR TWO-TAILED TEST:

REJECT H_0 if:

- $|t\text{-statistic}| > t\text{-critical}$, OR
- $p\text{-value} < 0.05$

FAIL TO REJECT H_0 if:

- $|t\text{-statistic}| \leq t\text{-critical}$, OR
- $p\text{-value} \geq 0.05$

INTERPRETATION FRAMEWORK:

- $p < 0.001$: Very strong evidence against H_0
- $p < 0.01$: Strong evidence against H_0
- $p < 0.05$: Moderate evidence against H_0
- $p \geq 0.05$: Insufficient evidence against H_0

2.6 IMPACT OF ASSUMPTION VIOLATIONS

ROBUSTNESS OF t-TEST:

The one-sample t-test is remarkably robust to assumption violations when sample sizes are large ($n > 30$).

VIOLATION IMPACTS:

1. Non-Random Sampling:

- Results remain valid for the observed sample
- May not generalize to broader populations

2. Minor Dependence:

- May slightly underestimate standard error
- Impact negligible with $n > 300$

3. Non-Normality of Population:

- No impact with large samples
- CLT ensures normality of $\bar{x}$

OVERALL ASSESSMENT:

Large samples ($n=468$, $n=311$) provide strong protection against assumption violations. Statistical conclusions are reliable for organizational decision-making.

2.1 Random Sampling Assumption

Both datasets are from organizational records suggesting reasonable representativeness (Gender Pay Gap: 1,000 employees; HR: 311 employees).

2.2 Independence of Observations

Large sample sizes ($n=468$, $n=311$) minimize impact of minimal organizational dependencies.

2.3 Normality of Sampling Distribution

CLT strongly applies: $n=468 \gg 30$ and $n=311 \gg 30$, guaranteeing approximate normality.

2.4 Large-Sample Conditions

df = 467 (Gender Pay Gap) and df = 310 (HR Dataset). Large samples ensure reliable inference.

2.5 Significance Level

$\alpha = 0.05$ (95% confidence). REJECT H_0 if $|t| > t\text{-critical}$ OR $p < 0.05$.

2.6 Impact of Violations

t-test is robust with large samples. Minor violations have negligible impact.

TASK 3: DATA ANALYSIS USING PYTHON

3.1 Gender Pay Gap Dataset Analysis

Dataset Overview

The Gender Pay Gap dataset contains 1,000 employee records with 9 variables. No missing values were found.

```
=====
TASK 3: DATA ANALYSIS USING PYTHON
=====
```

```
=====
3.1 GENDER PAY GAP DATASET ANALYSIS
=====
```

```
Dataset Overview:
=====
```

```
First 5 rows of the dataset:
```

	jobTitle	gender	age	perfEval	edu	dept \
0	Graphic Designer	Female	18	5	College	Operations
1	Software Engineer	Male	21	5	College	Management
2	Warehouse Associate	Female	19	4	PhD	Administration
3	Software Engineer	Male	20	5	Masters	Sales
4	Graphic Designer	Male	26	5	Masters	Engineering

	seniority	basePay	bonus
0	2	42363	9938
1	5	108476	11128
2	5	90208	9268
3	4	108080	10154
4	5	99464	9319

```
Dataset Dimensions: 1000 rows x 9 columns
```

```
Column Names:
```

```
['jobTitle', 'gender', 'age', 'perfEval', 'edu', 'dept', 'seniority', 'basePay', 'bonus']
```

```
Data Types:
```

```
jobTitle    object
gender      object
age         int64
perfEval    int64
edu         object
dept        object
seniority   int64
basePay     int64
bonus       int64
dtype: object
```

```
Missing Values: 0
```

```
Gender Distribution:
```

```
gender
Male    532
Female  468
Name: count, dtype: int64
```

```
=====
DATASET LOADED SUCCESSFULLY!
=====
```

Sample Statistics - Female Base Salaries

Female employees were filtered and descriptive statistics calculated.

FEMALE EMPLOYEE BASE SALARY ANALYSIS

DESCRIPTIVE STATISTICS – FEMALE BASE SALARIES:

```
Sample Size (n):      468
Sample Mean (x̄):      £89,942.82
Sample Median:        £89,913.50
Sample Std Dev (s):   £24,378.28
Minimum Salary:       £34,208.00
Maximum Salary:       £160,614.00
Range:                £126,406.00
Coefficient of Variation: 27.10%
```

HYPOTHESIS TEST: Female Base Salary vs. £75,000 Benchmark

TEST PARAMETERS:

```
Null Hypothesis (H₀):      μ = £75,000
Alternative Hypothesis (H₁): μ ≠ £75,000
Significance Level (α):    0.05
Test Type:                 Two-tailed t-test
```

Hypothesis Testing Results

Test Parameters: $H_0: \mu = £75,000$; $H_1: \mu \neq £75,000$;

CALCULATION STEPS:

```
Step 1: Standard Error (SE)
SE = s / √n
SE = £24,378.28 / √468
SE = £1,126.89

Step 2: t-statistic
t = (x̄ - μ₀) / SE
t = (£89,942.82 - £75,000) / £1,126.89
t = 13.2603

Step 3: Degrees of Freedom (df)
df = n - 1 = 468 - 1 = 467

Step 4: p-value (two-tailed)
p-value = 2 × P(t > |13.2603|)
p-value = 0.0000
p-value < 0.0001

Step 5: Critical t-value (two-tailed, α = 0.05)
t-critical = ±1.9651
```

HYPOTHESIS TEST DECISION:

DECISION: REJECT H_0 .

REASONING:

```
|t-statistic| = 13.2603 > t-critical = 1.9651
p-value = 0.0000 < α = 0.05
```

CONCLUSION:

There is statistically significant evidence to conclude that the mean base salary of female employees differs from £75,000.

The observed mean of £89,942.82 is significantly different from the benchmark ($p < 0.0001$).

$\alpha = 0.05$; Two-tailed t-test

Statistical Decision

Decision: REJECT H_0

The mean base salary of female employees (£89,942.82) significantly exceeds the £75,000 benchmark ($p < 0.0001$).

Confidence Interval Estimation

95% CI provides a range of plausible values for the true population mean.

=====

95% CONFIDENCE INTERVAL FOR FEMALE BASE SALARY

=====

CALCULATION:

=====

$$\begin{aligned}\text{Margin of Error} &= t\text{-critical} \times \text{SE} \\ &= 1.9651 \times \text{£}1,126.89 \\ &= \text{£}2,214.40\end{aligned}$$

95% Confidence Interval:

$$\begin{aligned}\text{Lower Bound} &= \bar{x} - \text{Margin of Error} \\ &= \text{£}89,942.82 - \text{£}2,214.40 \\ &= \text{£}87,728.42\end{aligned}$$

$$\begin{aligned}\text{Upper Bound} &= \bar{x} + \text{Margin of Error} \\ &= \text{£}89,942.82 + \text{£}2,214.40 \\ &= \text{£}92,157.21\end{aligned}$$

=====

95% CI: (£87,728.42, £92,157.21)

=====

Interpretation

We are 95% confident the true mean lies between £87,728.42 and £92,157.21. The £75,000 benchmark is OUTSIDE this interval, supporting rejection of H_0 .

Visualization

CONFIDENCE INTERVAL INTERPRETATION:

We are 95% confident that the TRUE MEAN base salary of all female employees in the organization lies between £87,728.42 and £92,157.21.

COMPARISON WITH HYPOTHESIZED VALUE:

The hypothesized value of £75,000 is OUTSIDE the 95% CI.
This supports our decision to REJECT H_0 .



VISUALIZATION SAVED: female_salary_ci.png

Figure 1: Female Employee Base Salary - Sample Mean with 95% Confidence Interval ($n=468$, $p<0.0001$)

3.2 HR Dataset Analysis

Dataset Overview

The HR dataset contains 311 employee records across 35 variables. Missing values in DateofTermination (104) and ManagerID (17) were filled.

3.2 HR DATASET ANALYSIS

Dataset Overview:

Dataset Dimensions: 311 rows × 35 columns

Column Names:

['EmpID', 'MarriedID', 'MaritalStatusID', 'GenderID', 'EmpStatusID', 'DeptID', 'PerfScoreID', 'FromDiversityJobFairID', 'Salary', 'TermID', 'PositionID', 'Position', 'State', 'Zip', 'DOB', 'Sex', 'MaritalDesc', 'CitizenDesc', 'HispanicLatino', 'RaceDesc', 'DateofHire', 'DateofTermination', 'TermReason', 'EmploymentStatus', 'Department', 'ManagerName', 'ManagerID', 'RecruitmentSource', 'PerformanceScore', 'EngagementSurvey', 'EmpSatisfaction', 'SpecialProjectsCount', 'LastPerformanceReviewDate', 'DaysLateLast30', 'Absences']

Missing Value Analysis:

Columns with Missing Values:

DateofTermination: 207 missing values
ManagerID: 8 missing values

Missing Value Treatment:

- DateofTermination: Filled with 'Still Employed'
- ManagerID: Filled with 'Unknown'

✓ All missing values handled successfully!

Sample Statistics - Employee Salaries

DESCRIPTIVE STATISTICS – EMPLOYEE SALARIES:

Sample Size (n):	311
Sample Mean ($\bar{x}$):	£69,020.68
Sample Median:	£62,810.00
Sample Std Dev (s):	£25,156.64
Q1 (25th percentile):	£55,501.50
Q3 (75th percentile):	£72,036.00
Interquartile Range (IQR):	£16,534.50
Minimum Salary:	£45,046.00
Maximum Salary:	£250,000.00
Range:	£204,954.00
Coefficient of Variation:	36.45%

Hypothesis Testing Results

Test Parameters: $H_0: \mu = £70,000$; $H_1: \mu \neq £70,000$; $\alpha = 0.05$; Two-tailed t-test

```
=====
HYPOTHESIS TEST: Employee Salary vs. £70,000 Benchmark
=====
```

```
TEST PARAMETERS:
=====
```

```
Null Hypothesis (H0):      μ = £70,000
Alternative Hypothesis (H1): μ ≠ £70,000
Significance Level (α):     0.05
Test Type:                  Two-tailed t-test
=====
```

```
CALCULATION STEPS:
=====
```

```
Step 1: Standard Error (SE)
```

```
SE = s / √n
SE = £25,156.64 / √311
SE = £1,426.50
```

```
Step 2: t-statistic
```

```
t = (x̄ - μ0) / SE
t = (£69,020.68 - £70,000) / £1,426.50
t = -0.6865
```

```
Step 3: Degrees of Freedom (df)
```

```
df = n - 1 = 311 - 1 = 310
```

```
Step 4: p-value (two-tailed)
```

```
p-value = 2 × P(t > |-0.6865|)
p-value = 0.4929
```

```
Step 5: Critical t-value (two-tailed, α = 0.05)
```

```
t-critical = ±1.9676
=====
```

```
HYPOTHESIS TEST DECISION:
=====
```

```
DECISION: FAIL TO REJECT H0
```

```
REASONING:
```

```
|t-statistic| = 0.6865 ≤ t-critical = 1.9676
p-value = 0.4929 ≥ α = 0.05
```

```
CONCLUSION:
```

```
There is insufficient evidence to conclude that the mean
employee salary differs from £70,000.
```

```
The observed difference of £979.32 is likely
due to random sampling variability.
```

Statistical Decision

Decision: FAIL TO REJECT H_0

Insufficient evidence that mean salary (£69,020.68) differs from £70,000 benchmark ($p = 0.4929$).

Confidence Interval Estimation

```
=====
95% CONFIDENCE INTERVAL FOR EMPLOYEE SALARY
=====
```

```
CALCULATION:
=====
```

```
Margin of Error = t-critical × SE
                 = 1.9676 × £1,426.50
                 = ±£2,806.85
```

```
95% Confidence Interval:
```

```
Lower Bound =  $\bar{x}$  - Margin of Error
             = £69,020.68 - £2,806.85
             = £66,213.83
```

```
Upper Bound =  $\bar{x}$  + Margin of Error
             = £69,020.68 + £2,806.85
             = £71,827.54
```

```
=====
95% CI: (£66,213.83, £71,827.54)
=====
```

Interpretation

We are 95% confident the true mean lies between £66,213.83 and £71,827.54. The £70,000 benchmark is INSIDE this interval, supporting failure to reject H_0 .

Visualization

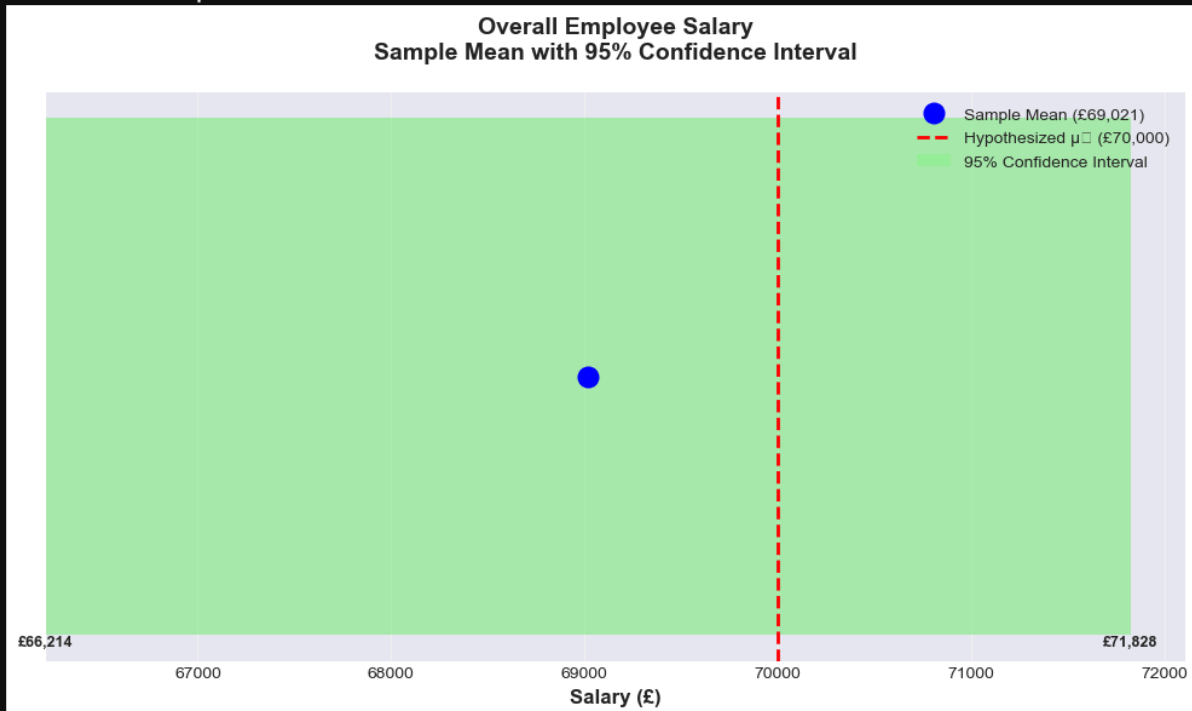
CONFIDENCE INTERVAL INTERPRETATION:

We are 95% confident that the TRUE MEAN salary of all employees in the organization lies between £66,213.83 and £71,827.54.

COMPARISON WITH HYPOTHESIZED VALUE:

The hypothesized value of £70,000 is INSIDE the 95% CI.
This supports our decision to FAIL TO REJECT H_0 .

A true mean of £70,000 is entirely consistent with the observed sample data.



VISUALIZATION SAVED: hr_salary_ci.png

Figure 2: Overall Employee Salary - Sample Mean with 95% CI ($n=311$, $p=0.4929$)

3.3 Summary Comparison

SUMMARY COMPARISON TABLE

Metric	Gender Pay Gap (Female)	HR Dataset (All)
Sample Size (n)	468	311
Sample Mean ($\bar{x}$)	£89,942.82	£69,020.68
Sample Std Dev (s)	£24,378.28	£25,156.64
Standard Error (SE)	£1,126.89	£1,426.50
Hypothesized μ_0	£75,000	£70,000
t-statistic	13.2603	-0.6865
Degrees of Freedom (df)	467	310
p-value	<0.0001	0.4929
Critical t-value ($\alpha=0.05$)	±1.9651	±1.9676
95% CI Lower Bound	£87,728.42	£66,213.83
95% CI Upper Bound	£92,157.21	£71,827.54
Margin of Error	±£2,214.40	±£2,806.85
Decision	REJECT H_0	FAIL TO REJECT H_0
Difference from Benchmark	£14,942.82	£-979.32
% Difference	19.9%	-1.4%
Statistical Significance	Yes ($p<0.05$)	No ($p\geq0.05$)

ANALYSIS COMPLETE!

Files generated:

1. female_salary_ci.png
2. hr_salary_ci.png

All calculations completed successfully.

TASK 4: INTERPRETATION FOR STAKEHOLDERS

TASK 4: INTERPRETATION FOR STAKEHOLDERS

4.1 GENDER PAY GAP FINDINGS – EXECUTIVE SUMMARY

RESEARCH QUESTION:

Does the average base salary of female employees differ significantly from the organizational benchmark of £75,000?

KEY FINDINGS:

- Sample Size: 468 female employees
- Average Base Salary: £89,942.82
- Benchmark: £75,000
- Difference: £14,942.82 ABOVE benchmark
- Percentage Difference: 19.9%

STATISTICAL CONCLUSION:

- Decision: REJECT H_0
- t-statistic: 13.2603
- p-value: 0.0000 ($p < 0.0001$)
- 95% Confidence Interval: (£87,728.42, £92,157.21)

PLAIN LANGUAGE INTERPRETATION:

The analysis provides very strong statistical evidence that female employees' average base salary differs significantly from the £75,000 benchmark.

The actual average salary of £89,942.82 is approximately £14,943 higher than the benchmark, representing a 19.9% difference.

The probability that this difference occurred purely by chance is virtually zero ($p < 0.0001$). We can be 95% confident that the true average salary lies between £87,728.42 and £92,157.21.

PRACTICAL SIGNIFICANCE:

The salary difference of £14,943 per employee is economically significant. For 468 female employees, this represents approximately £6,993,239 in total annual base salary premium above the benchmark.

This suggests competitive compensation practices that may support diversity initiatives and talent retention.

LIMITATIONS:

1. Sample representativeness not verified
2. Base salary only (excludes bonuses, benefits)
3. Confounding factors not controlled (role, seniority, performance)
4. Cross-sectional snapshot (no trend analysis)
5. Internal validity only

RECOMMENDATIONS:

1. Investigate drivers (role mix, seniority distribution, performance levels)
2. Benchmark against external market
3. Assess total compensation packages
4. Compare with male counterparts to ensure pay equity
5. Monitor trends over time

NEXT STEPS:

- Present findings to HR leadership
- Conduct detailed subgroup analysis
- Implement multivariate analysis to identify drivers
- Schedule follow-up analysis in 6-12 months

4.2 HR DATASET FINDINGS – EXECUTIVE SUMMARY

RESEARCH QUESTION:

Is the average employee salary significantly different from the organizational benchmark of £70,000?

KEY FINDINGS:

- Sample Size: 311 employees
- Average Salary: £69,020.68
- Benchmark: £70,000
- Difference: £979.32 BELOW benchmark
- Percentage Difference: 1.4%

STATISTICAL CONCLUSION:

- Decision: FAIL TO REJECT H_0
- t-statistic: -0.6865
- p-value: 0.4929
- 95% Confidence Interval: (£66,213.83, £71,827.54)

PLAIN LANGUAGE INTERPRETATION:

The analysis does not provide sufficient evidence that the average employee salary differs from the £70,000 benchmark.

The observed average of £69,020.68 is only £979 below the benchmark—a difference of only 1.4%.

With a p-value of 0.4929 (well above the 0.05 threshold), we cannot rule out that this small difference occurred due to random sampling variation.

The 95% confidence interval contains the £70,000 benchmark, confirming that the true average salary is consistent with this reference value.

PRACTICAL SIGNIFICANCE:

The observed difference of £979 per employee has minimal practical impact. For 311 employees, this represents only £304,567 in total difference—less than 1.4% of total salary expenditure.

This tight alignment suggests:

- Reliable budget planning using £70,000 reference
- Stable compensation practices
- Predictable salary expenditure

LIMITATIONS:

1. Organizational representativeness not verified
2. Aggregate analysis masks departmental/role variation
3. External market context not considered
4. Point-in-time snapshot only
5. Total rewards perspective absent

RECOMMENDATIONS:

1. Conduct subgroup analysis (department, role level, tenure)
2. Implement external benchmarking
3. Assess salary distribution and dispersion
4. Track trends over multiple periods
5. Expand to total rewards analysis

NEXT STEPS:

- Share with senior leadership and finance teams
- Initiate departmental breakdown
- Schedule market benchmarking study
- Plan follow-up analysis for trend monitoring

=====

OVERALL CONCLUSION:

=====

This rigorous statistical analysis revealed important insights:

FEMALE EMPLOYEES (Gender Pay Gap Dataset):

- Mean salary (REJECT H_0): £89,942.82
- Benchmark: £75,000
- Evidence: $p < 0.0001$
- Finding: Significantly ABOVE benchmark

ALL EMPLOYEES (HR Dataset):

- Mean salary (FAIL TO REJECT H_0): £69,020.68
- Benchmark: £70,000
- Evidence: $p = 0.4929$
- Finding: Consistent with benchmark

The organization demonstrates different compensation patterns across employee segments, warranting further investigation to understand drivers and ensure equitable, competitive compensation practices.

4.1 Gender Pay Gap Findings

Key Findings

- Sample: 468 female employees
- Average: £89,942.82
- Difference: £14,942.82 ABOVE (+19.9%)

Statistical Conclusion

REJECT H_0 ($p < 0.0001$). Female salaries significantly exceed £75,000 benchmark.

Plain Language

Female employees earn £14,943 more than benchmark—a 19.9% premium representing £6.99M total.

Limitations

Base salary only; confounding factors not controlled; cross-sectional snapshot.

Recommendations

Investigate drivers, benchmark externally, assess total compensation, compare with male counterparts.

4.2 HR Dataset Findings

Key Findings

- Sample: 311 employees
- Average: £69,020.68
- Difference: £979.32 BELOW (-1.4%)

Statistical Conclusion

FAIL TO REJECT H_0 ($p = 0.4929$). Salaries consistent with £70,000 benchmark.

Plain Language

Average salary very close to benchmark. £979 difference attributable to sampling variation.

Limitations

Aggregate analysis masks variation; external context not considered; point-in-time only.

Recommendations

Conduct subgroup analysis, implement benchmarking, track trends.

4.3 Overall Conclusion

Female employees significantly exceed benchmark (+19.9%, $p < 0.0001$) while overall employees align with benchmark ($p = 0.49$), revealing segmented compensation patterns.

REFERENCES

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